

Universidad Internacional de Valencia (VIU)

Master's Degree in Data Science and Big Data
Advanced Statistics



Programming in RStudio

Activity 1 – Data Analysis Using Regression Models

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INDEX

1. Introducción	3
1.1 Contexto y motivación	3
1.2 Descripción de la actividad	3
1.3 Objetivos del análisis	3
2. Descripción de los datos	4
2.1 Origen y estructura	4
2.2 Preparación de los datos	4
2.3 Análisis exploratorio	4
3. Regresión lineal y polinómica	5
3.1 Selección del grado	5
3.2 Diagnóstico y transformación	7
4. Regresión multilínea	8
4.1 Estimación y selección de variables	8
4.2 Interpretación de los efectos	9
4.3 Diagnóstico y validación	10
5. Regresión logística	11
5.1 Estimación y bondad de ajuste	11
5.2 Odds ratios	11
5.3 Capacidad predictiva y umbral	12
6. Conclusiones	14
7. Limitaciones y trabajo futuro	14
Anexo. Código R	16

1. Introduction

1.1 Context and Motivation

Wages are one of the most studied indicators in labour economics, as they reflect the value the market assigns to each worker's characteristics: their education, their age or the industry in which they work. Identifying which factors are associated with higher wages is useful both for individuals, when making decisions about their education and career, and for policymakers.

In the United States, moreover, access to health insurance largely depends on employment, since in most cases it is provided by the employer, so identifying who remains uncovered has direct public health implications. From a statistical point of view, both questions are regression problems: one with a continuous response and the other with a binary response. This makes them an ideal case for applying the three families of models covered in this subject to a single dataset.

1.2 Activity Description

The activity consists of carrying out a complete statistical analysis on a dataset, applying linear and polynomial regression, multiple linear regression and logistic regression. For each model, the parameters are estimated, the assumptions are validated and interpretable conclusions are drawn. The analysis was developed entirely in R 4.6.1 on RStudio, using the packages *ISLR2* (data), *ggplot2* and *GGally* (visualisation), *car* and *lmtest* (model diagnostics), *sandwich* (robust standard errors), *caret* (data partitioning and metrics) and *pROC* (ROC curves). The complete code is included in the Annex.

1.3 Analysis Objectives

1. Describe the dataset statistically and prepare the variables for analysis.
2. Model the relationship between age and wage, determining whether it is linear or requires a polynomial term.
3. Quantify the joint effect of sociodemographic and occupational variables on wages through multiple linear regression.
4. Estimate the probability of holding health insurance through logistic regression and assess its predictive performance.
5. Validate the assumptions of each model and its ability to generalise to data not used in fitting..

2. Data Description

2.1 Source and Structure

The *Wage* dataset was used, included in R's *ISLR2* package and sourced from the *Current Population Survey*. It contains information on 3,000 male workers from the *Middle Atlantic* region of the United States, collected between 2003 and 2009. This dataset was chosen because it combines, within a single sample, a continuous response variable (wage) and a binary one (health insurance), which allows the three required families of models to be applied to real data from a reliable source.

The dataset consists of 11 variables: four numerical (year of record, age, annual wage in thousands of dollars and its logarithm) and seven categorical, most notably education level, with five ordered levels ranging from not having completed high school to holding an advanced degree, job class, marital status, health status and health insurance.

```
> str(Wage)
'data.frame':  3000 obs. of  11 variables:
 $ year      : int  2006 2004 2003 2003 2005 2008 2009 2008 2006 2004 ...
 $ age       : int  18 24 45 43 50 54 44 30 41 52 ...
 $ maritl    : Factor w/ 5 levels "1. Never Married",...: 1 1 2 2 4 2 2 1 1 2 ...
 $ race      : Factor w/ 4 levels "1. White", "2. Black",...: 1 1 1 3 1 1 4 3 2 1 ...
 $ education : Factor w/ 5 levels "1. < HS Grad",...: 1 4 3 4 2 4 3 3 3 2 ...
 $ region    : Factor w/ 9 levels "1. New England",...: 2 2 2 2 2 2 2 2 2 2 ...
 $ jobclass  : Factor w/ 2 levels "1. Industrial",...: 1 2 1 2 2 2 1 2 2 2 ...
 $ health    : Factor w/ 2 levels "1. <=Good", "2. >=Very Good": 1 2 1 2 1 2 2 1 2 2 ...
 $ health_ins: Factor w/ 2 levels "1. Yes", "2. No": 2 2 1 1 1 1 1 1 1 1 ...
 $ logwage   : num  4.32 4.26 4.88 5.04 4.32 ...
 $ wage      : num  75 70.5 131 154.7 75 ...
```

Figure 1. Structure of the Wage dataset.

2.2 Data Preparation

The check for missing values found none in any variable, so no imputation techniques were required. Three transformations were applied to the original dataset.

First, the variable *region* was removed, since all 3,000 observations belong to the same geographical region and, being constant, it provides no information to the model. Second, *logwage* was discarded, as it is a direct transformation of the response variable: using it as a predictor would cause data leakage and produce an artificially perfect fit. When the logarithm is needed, it will be computed directly within the model formula. Finally, the binary variable *ins* was created, taking the value 1 if the worker holds health insurance and 0 otherwise, which will act as the response of the logistic model.

This last variable shows a moderate imbalance: 69.4% of workers are insured, compared with 30.6% who are not. This will be taken into account when evaluating the performance of the classification model, as it affects the interpretation of overall accuracy.

2.3 Exploratory Analysis

The wage distribution shows marked positive skewness and a distinct group of observations above 250 thousand dollars (Figure 2a), which anticipates normality problems in the residuals. The relationship between age and wage is not linear (Figure 2b): the LOESS smoother shows a steep increase up to ages 45–50, followed by a

levelling off and a slight decline, describing an inverted U shape characteristic of the working life cycle. This pattern justifies exploring polynomial terms, since a straight line cannot capture this change in trend.

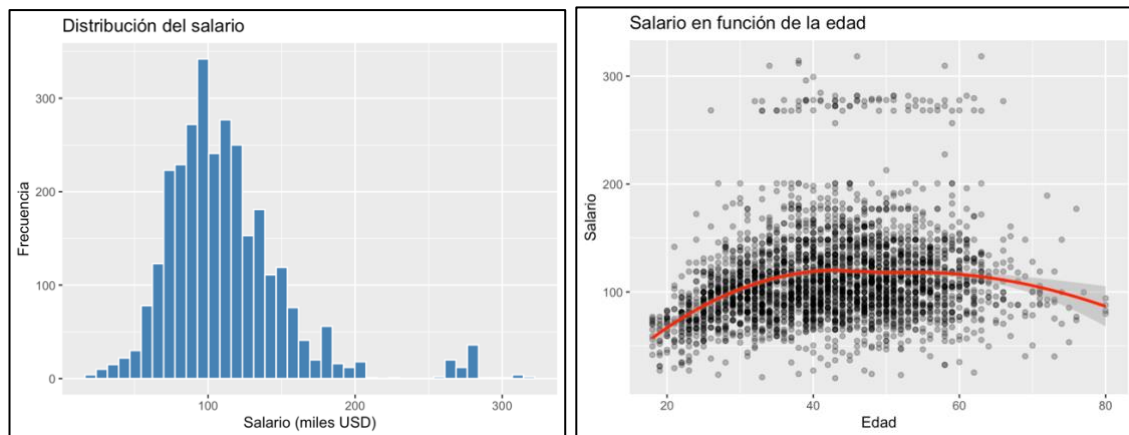


Figure 2. (a) Wage distribution. (b) Wage versus age with LOESS smoothing.

Regarding the categorical variables, wages increase monotonically with education level and are higher in the *Information* sector than in the *Industrial* sector (Figure 3). Both patterns anticipate the relevance of these variables in the multiple linear model. Correlations between the numerical variables are weak, ruling out collinearity problems among them.

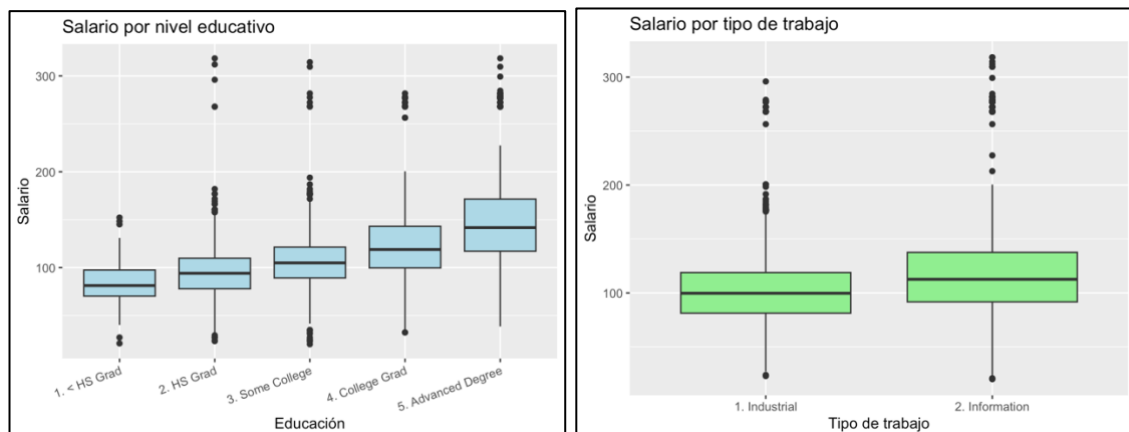


Figure 3. Wage by education level and by job class.

3. Linear and Polynomial Regression

3.1 Degree Selection

A simple linear model was first estimated, with wage as the response and age as the only predictor. The coefficient was highly significant ($p < 2e-16$) and indicates that each additional year is associated with an average increase of 707 dollars in annual wage. However, the R^2 is only 0.038: age alone explains less than 4% of wage variability, and the residual standard error amounts to 40.93 thousand dollars.

```

Call:
lm(formula = wage ~ age, data = datos)

Residuals:
    Min       1Q   Median       3Q      Max
-100.265  -25.115   -6.063   16.601   205.748

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  81.70474    2.84624   28.71  <2e-16 ***
age           0.70728    0.06475   10.92  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.93 on 2998 degrees of freedom
Multiple R-squared:  0.03827,    Adjusted R-squared:  0.03795
F-statistic: 119.3 on 1 and 2998 DF,  p-value: < 2.2e-16

```

Figure 4. Results of the simple linear model.

Given the non-linear pattern observed, polynomial models of degrees 2 to 4 were estimated using orthogonal polynomials, which avoid multicollinearity between successive powers of age. The comparison between nested models through an F-test shows significant improvements from degree 1 to 2 ($F = 143.60$; $p < 2e-16$) and from 2 to 3 ($F = 9.89$; $p = 0.002$), but not from 3 to 4 ($p = 0.051$). The comparison table reinforces this conclusion, as the adjusted R^2 improves by only 0.0009 when the fourth degree is added and the AIC difference is below 2, the value under which two models are considered equivalent. Applying the principle of parsimony, the third-degree polynomial is selected.

```

> anova(m1, m2, m3, m4)
Analysis of Variance Table

Model 1: wage ~ age
Model 2: wage ~ poly(age, 2)
Model 3: wage ~ poly(age, 3)
Model 4: wage ~ poly(age, 4)
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1    2998 5022216
2    2997 4793430   1    228786 143.6025 < 2.2e-16 ***
3    2996 4777674   1    15756   9.8894 0.001679 **
4    2995 4771604   1     6070   3.8101 0.051039 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

> comparacion
  grado  R2_adj  AIC
1     1 0.03795313 30788.67
2     2 0.08147259 30650.80
3     3 0.08418615 30642.92
4     4 0.08504433 30641.11

```

Figure 5. F-test between nested models and comparison using adjusted R^2 and AIC.

The selected model reaches an adjusted R^2 of 0.084, more than double that of the linear model, and reduces the residual error to 39.93. According to its estimates, wages peak at age 47, at 120.7 thousand dollars. Point predictions illustrate the life-cycle profile: 87.8 at age 25, with a 95% confidence interval of [84.6; 90.9]; 120.5 at age 45; and 110.5 at age 65. Figure 6 shows that the curves for degrees 2, 3 and 4 are practically indistinguishable from each other, whereas the straight line fails to capture the decline in the final years of working life.

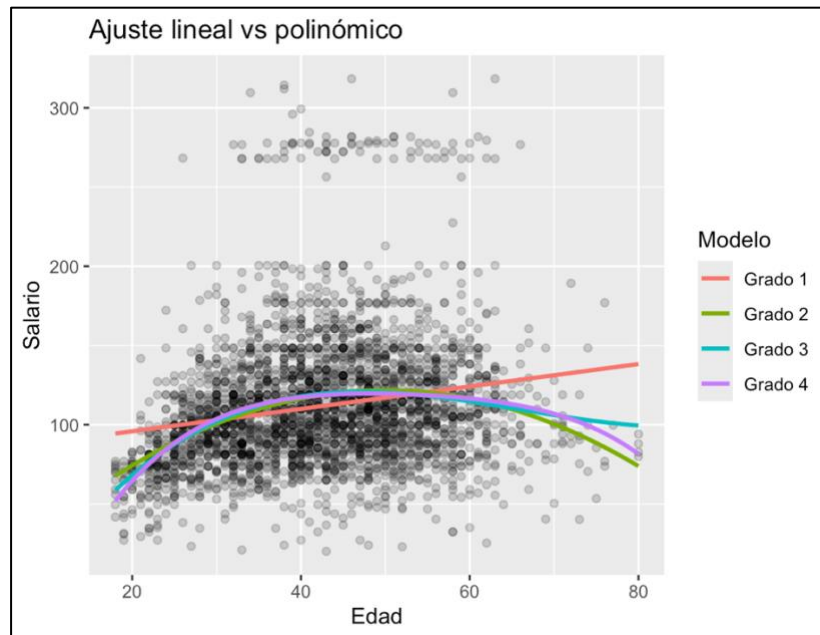


Figura 6. Comparación del ajuste lineal frente al polinómico.

3.2 Diagnostics and Transformation

Graphical diagnostics of the third-degree model show a flat residual line, confirming that the functional form is adequate, but reveal two problems: dispersion increases with fitted values, a sign of heteroscedasticity, and the Q-Q plot deviates sharply in the right tail, corresponding to the group of highest wages. Formal tests confirm both diagnoses, rejecting normality (Shapiro-Wilk: $W = 0.876$; $p < 2.2e-16$) and homoscedasticity (Breusch-Pagan: $BP = 35.47$; $p = 9.7e-08$). In contrast, no observation exceeds a Cook's distance of 0.5, so there are no influential points.

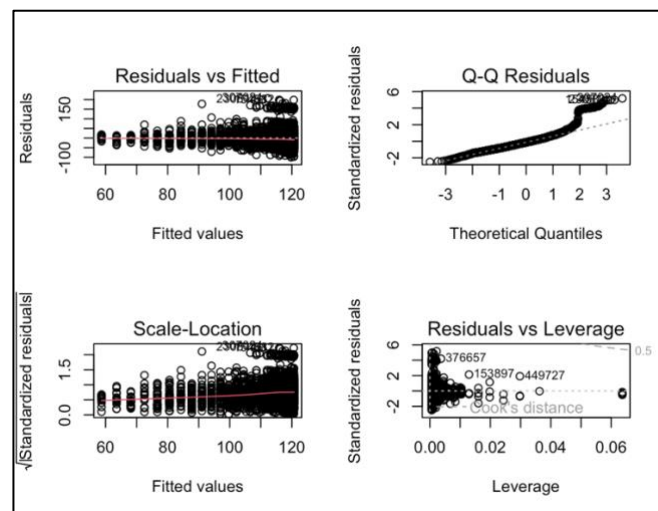


Figure 7. Residual diagnostics of the third-degree polynomial model.

To correct these problems, the same model was estimated applying a logarithmic transformation to the response variable. The improvement is notable: the Shapiro-Wilk statistic rises from 0.876 to 0.974 and the Breusch-Pagan statistic falls from 35.47 to 11.28 ($p = 0.010$), so heteroscedasticity is no longer significant at the 1% level.

Both assumptions are still formally rejected, but this result should be qualified. With a sample size of 3,000 observations, the Shapiro-Wilk test detects minimal deviations from normality, and the central limit theorem ensures the asymptotic validity of inference on the coefficients. It should also be noted that the R^2 values of both models are not directly comparable, as the response variable is expressed on different scales.

Shapiro-Wilk normality test data: residuals(m3) W = 0.87556, p-value < 2.2e-16 > bptest(m3) # homoced studentized Breusch-Pagan test data: m3 BP = 35.471, df = 3, p-value = 9.689e-08	Shapiro-Wilk normality test data: residuals(m3_log) W = 0.97359, p-value < 2.2e-16 > bptest(m3_log) studentized Breusch-Pagan test data: m3_log BP = 11.276, df = 3, p-value = 0.01032
---	---

Figure 8. Assumption tests before and after the logarithmic transformation.

For this reason, the models in the following sections use the logarithm of wage as the response variable. This choice also offers an interpretive advantage: the coefficients can be translated directly into percentage changes in wage.

4. Multiple Linear Regression

4.1 Estimation and Variable Selection

A model was estimated with log wage as the response and eight predictors: the third-degree polynomial of age, year of record, marital status, race, education level, job class, health status and health insurance.

The multicollinearity diagnostic rules out any problem. Since categorical variables are included, R provides the generalised variance inflation factor; all $GVIF^{1/(2 \cdot Df)}$ values lie between 1.00 and 1.08, well below the usual threshold of 2.2, equivalent to a VIF of 5.

	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$
poly(age, 3)	1.571097	3	1.078203
year	1.007255	1	1.003621
maritl	1.491922	4	1.051280
race	1.098816	3	1.015830
education	1.265274	4	1.029848
jobclass	1.137029	1	1.066316
health	1.070577	1	1.034687
health_ins	1.101122	1	1.049344

Figure 9. Generalised variance inflation factors.

Stepwise variable selection by AIC in both directions retained all eight predictors, indicating that all of them contribute relevant information to the model, even though some individual categories are not significant on their own. The model reaches an adjusted R^2 of 0.388, far above the 0.113 of the model including only age, and the F-test comparison between the two is highly significant ($F = 90.56$; $p < 2.2e-16$).

```
> summary(mn_full)

Call:
lm(formula = log(wage) ~ poly(age, 3) + year + maritl + race +
    education + jobclass + health + health_ins, data = datos)

Residuals:
    Min       1Q   Median       3Q      Max
-1.5665 -0.1452  0.0054  0.1543  1.2201

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -21.740869   4.993399  -4.354 1.38e-05 ***
poly(age, 3)1    2.085417   0.323041   6.456 1.25e-10 ***
poly(age, 3)2   -2.545481   0.294039  -8.657 < 2e-16 ***
poly(age, 3)3    0.739170   0.279543   2.644  0.00823 **
year           0.013027   0.002489   5.233 1.78e-07 ***
maritl2. Married  0.121464   0.014762   8.228 2.80e-16 ***
maritl3. Widowed  0.027375   0.064900   0.422  0.67320
maritl4. Divorced  0.002898   0.023876   0.121  0.90339
maritl5. Separated  0.076802   0.039563   1.941  0.05232
race2. Black     -0.037231   0.017382  -2.142  0.03228 *
race3. Asian     -0.022510   0.021085  -1.068  0.28580
race4. Other     -0.056329   0.045870  -1.228  0.21954
education2. HS Grad  0.080552   0.019184   4.199 2.76e-05 ***
education3. Some College  0.180012   0.020403   8.823 < 2e-16 ***
education4. College Grad  0.275324   0.020644  13.337 < 2e-16 ***
education5. Advanced Degree  0.424435   0.022779  18.633 < 2e-16 ***
jobclass2. Information  0.024596   0.010721   2.294  0.02185 *
health2. >=Very Good  0.056477   0.011506   4.909 9.66e-07 ***
health_ins2. No   -0.181134   0.011446  -15.826 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2752 on 2981 degrees of freedom
Multiple R-squared:  0.3915,    Adjusted R-squared:  0.3878
F-statistic: 106.5 on 18 and 2981 DF, p-value: < 2.2e-16
```

Figure 10. Results of the estimated multiple linear model.

4.2 Interpretation of Effects

As the response is on a logarithmic scale, the transformation $(e^{\beta} - 1) \cdot 100$ expresses each coefficient as a percentage change in wage, holding all other variables constant.

Education level emerges as the most influential factor and shows a clearly increasing gradient relative to the reference category, made up of those who did not complete high school: 8.4% more with a high school diploma, 19.7% with some college, 31.7% with a college degree and 52.9% with an advanced degree. Next in magnitude come the absence of health insurance, associated with a wage 16.6% lower; marital status, with an increase of 12.9% for married workers; very good health, with 5.8%; the *Information* sector, with 2.5%; and a time trend of approximately 1.3% per year over the period analysed.

The coefficients of the age polynomial cannot be interpreted directly, as they are orthogonal polynomials; their joint effect is the one described in the previous section.

(Intercept)	poly(age, 3)1
-100.0	704.8
poly(age, 3)2	poly(age, 3)3
-92.2	109.4
year	maritl2. Married
1.3	12.9
maritl3. Widowed	maritl4. Divorced
2.8	0.3
maritl5. Separated	race2. Black
8.0	-3.7
race3. Asian	race4. Other
-2.2	-5.5
education2. HS Grad	education3. Some College
8.4	19.7
education4. College Grad	education5. Advanced Degree
31.7	52.9
jobclass2. Information	health2. >=Very Good
2.5	5.8
health_ins2. No	
-16.6	

Figure 11. Estimated effects on wage expressed as percentages.

4.3 Diagnóstico y validación

Graphical diagnostics improve noticeably compared with the simple model: the residual cloud shows no structure and the upper band corresponding to the highest wages has disappeared, now explained by education level and the remaining predictors. Heteroscedasticity (BP = 84.87; $p = 1.19\text{e-}10$) and the rejection of normality ($W = 0.964$) nevertheless persist, although the latter with a statistic considerably closer to one. The influence analysis rules out problematic observations, as the maximum Cook's distance is 0.0124, far below any threshold of concern.

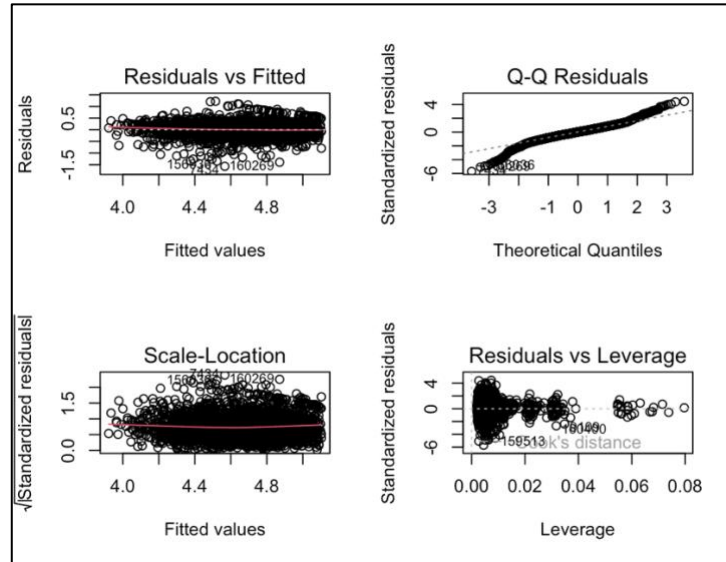


Figure 12. Residual diagnostics of the multiple linear model.

Given the heteroscedasticity detected, standard errors were recalculated using the HC3 robust estimator, which provides valid inference without assuming constant variance. All conclusions hold and only the *Separated* category becomes significant at the 5% level ($p = 0.011$ versus 0.052), confirming the robustness of the results.

Finally, the model's predictive performance was validated through a random split into 70% training and 30% test data. On the test set, after back-transforming the logarithm to express predictions on the original scale, the RMSE is 33.70 and the mean absolute error 22.38 thousand dollars, with an R^2 of 0.338. The proximity between this value and the 0.388 obtained in fitting indicates that the model does not overfit and generalises correctly to data not used in estimation.

RMSE	MAE	R2_test
33.7047217	22.3780325	0.3381854

Figure 13. Validation metrics on the test set.

5. Logistic Regression

5.1 Estimation and Goodness of Fit

The final model estimates the probability that a worker holds health insurance. The data were split into 70% training, with 2,101 observations, and 30% test, with 899. The split was stratified to preserve the proportion of the response variable, which remains at 30.6% uninsured and 69.4% insured in both subsets.

A full model with eight predictors was fitted, from which AIC selection removed four (year, marital status, race and health status), reducing the AIC from 2,332.6 to 2,325.4 with virtually no change in residual deviance. The final model includes age, education level, job class and wage, and significantly improves on the null model according to the likelihood ratio test ($\chi^2 = 276.94$; 7 df; $p < 2.2e-16$). McFadden's R^2 is 0.107, a modest fit indicating that the available variables only partially explain who holds coverage. Variance inflation factors, all below 1.12, rule out multicollinearity.

```
Call:
glm(formula = ins_f ~ age + education + jobclass + wage, family = binomial,
    data = train2)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.581916   0.277336  -9.310 < 2e-16 ***
age           0.012854   0.004453   2.887 0.00389 **
education2. HS Grad  0.365202  0.175536  2.080 0.03748 **
education3. Some College 0.651722  0.193351  3.371 0.00075 ***
education4. College Grad 0.429764  0.200187  2.147 0.03181 *
education5. Advanced Degree 0.209774  0.239163  0.877 0.38042
jobclass2. Information 0.326280  0.106071  3.076 0.00210 **
wage          0.022287   0.002023 11.015 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2586.4 on 2100 degrees of freedom
Residual deviance: 2309.4 on 2093 degrees of freedom
AIC: 2325.4

Number of Fisher Scoring iterations: 5
```

Figure 14. Logistic model selected by AIC.

```
Analysis of Deviance Table

Model 1: ins_f ~ 1
Model 2: ins_f ~ age + education + jobclass + wage
Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      2100      2586.4
2      2093      2309.4  7    276.94 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> c(AIC_null = AIC(lg_null),
+   AIC_step = AIC(lg_step),
+   R2_McFadden = 1 - lg_step$deviance / lg_step$null.deviance)
AIC_null AIC_step R2_McFadden
2588.3693904 2325.4318865 0.1070758
```

Figure 15. Likelihood ratio test and goodness-of-fit measures.

5.2 Odds Ratios

The coefficients of a logistic model are interpreted through odds ratios, obtained by exponentiating the estimates. A value greater than one indicates that the variable increases the odds of belonging to the positive class.

Wage is the most significant variable in the model ($z = 11.02$): each additional 1,000 dollars multiplies the odds of holding insurance by 1.023, equivalent to an increase of roughly 25% per 10,000 dollars. Age shows an effect of 1.013 per year, around 14% per decade. Working in the *Information* sector increases the odds by 39% relative to the industrial sector. Education level is significant in all its categories except advanced degree, whose confidence interval contains the value 1.

It is worth noting that, unlike in the multiple linear model, the effect of education does not follow an increasing pattern here. The most plausible explanation is that wage, included as a predictor, absorbs much of the effect of education: higher education leads to higher wages, and higher wages to a greater probability of coverage. Once wage is controlled for, the residual effect of education is smaller and loses its ordering.

	OR	2.5 %	97.5 %
(Intercept)	0.076	0.044	0.130
age	1.013	1.004	1.022
education2. HS Grad	1.441	1.021	2.032
education3. Some College	1.919	1.314	2.803
education4. College Grad	1.537	1.038	2.275
education5. Advanced Degree	1.233	0.772	1.971
jobclass2. Information	1.386	1.126	1.706
wage	1.023	1.018	1.027

Figure 16. Odds ratios with 95% confidence intervals.

5.3 Predictive Performance and Threshold

Applying the conventional 0.5 threshold on the test set, the model achieves an accuracy of 75.6%, significantly above the no information rate of 69.4% ($p = 2.08e-05$). However, the confusion matrix reveals markedly asymmetric behaviour: sensitivity reaches 93.1% versus a specificity of only 36.0%. The model correctly identifies almost all insured workers, but detects only one in three uninsured workers, a direct consequence of class imbalance. The Kappa index of 0.337 reflects moderate-to-low agreement once chance agreement is accounted for.

Confusion Matrix and Statistics		
	Reference	
Prediction	No	Si
No	99	43
Si	176	581
Accuracy : 0.7564		
95% CI : (0.727, 0.7841)		
No Information Rate : 0.6941		
P-Value [Acc > NIR] : 2.082e-05		
Kappa : 0.3366		
McNemar's Test P-Value : < 2.2e-16		
Sensitivity : 0.9311		
Specificity : 0.3600		
Pos Pred Value : 0.7675		
Neg Pred Value : 0.6972		
Prevalence : 0.6941		
Detection Rate : 0.6463		
Detection Prevalence : 0.8420		
Balanced Accuracy : 0.6455		
'Positive' Class : Si		

Figure 17. Confusion matrix with a 0.5 threshold.

The ROC curve allows the model's discriminant capacity to be assessed independently of the chosen threshold. The area under the curve is 0.752, with a 95% confidence interval of [0.717; 0.788], clearly above the 0.5 that would correspond to random classification. This therefore represents an acceptable discriminant capacity.

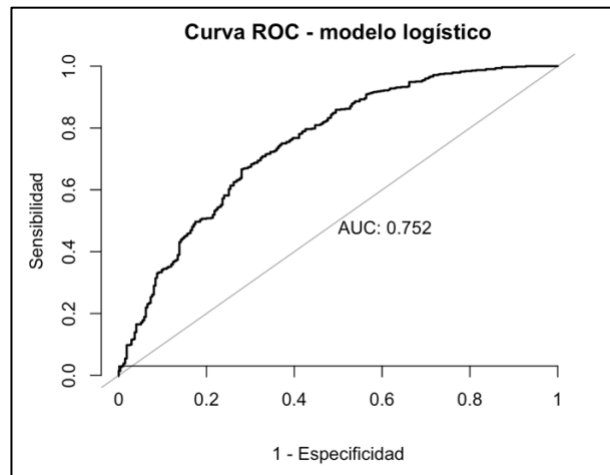


Figure 18. ROC curve of the logistic model on the test set.

Given the imbalance observed between sensitivity and specificity, the optimal threshold was determined using Youden's index, which yielded 0.694, a value practically identical to the prevalence of the positive class. With this threshold the model becomes balanced: specificity rises to 72.0% and sensitivity falls to 66.7%, so balanced accuracy improves from 64.6% to 69.3%. In exchange, overall accuracy drops to 68.3% and no longer exceeds the no information rate.

The choice between the two thresholds therefore depends on the objective of the analysis. If the aim is to detect uninsured workers, for instance for health policy purposes, Youden's threshold is preferable despite its lower overall accuracy. This contrast illustrates why accuracy is a misleading metric when classes are imbalanced.

Confusion Matrix and Statistics		
	Reference	
Prediction	No	Si
No	198	208
Si	77	416
Accuracy : 0.683		
95% CI : (0.6514, 0.7133)		
No Information Rate : 0.6941		
P-Value [Acc > NIR] : 0.777		
Kappa : 0.3412		
McNemar's Test P-Value : 1.355e-14		
Sensitivity : 0.6667		
Specificity : 0.7200		
Pos Pred Value : 0.8438		
Neg Pred Value : 0.4877		
Prevalence : 0.6941		
Detection Rate : 0.4627		
Detection Prevalence : 0.5484		
Balanced Accuracy : 0.6933		
'Positive' Class : Si		

Figure 19. Optimal threshold according to Youden's index and resulting confusion matrix.

6. Conclusions

The analysis has fulfilled the five objectives set out in the introduction.

Regarding the relationship between age and wage, it is confirmed that it is not linear. A third-degree polynomial doubles the fit of the linear model and places the wage peak at around age 47, reproducing the life-cycle profile described in the economic literature. Even so, age alone explains less than 9% of the variability, which justified extending the model.

The multiple linear model raises the proportion of explained variance to 38.8% and identifies education level as the main determinant of wages, with a difference of close to 53% between those holding an advanced degree and those who did not complete high school. Validation on data not used in fitting yields an R^2 of 0.338, very close to the training value, ruling out overfitting and supporting the model's ability to generalise.

Regarding health coverage, the logistic model achieves an AUC of 0.752 and identifies wage as the most decisive variable, followed by job class and age. The threshold analysis shows that, with imbalanced classes, optimising overall accuracy leads to a model that barely detects the minority class, which is precisely the most relevant one from a social standpoint.

From a methodological perspective, the work highlights the importance of assumption diagnostics. The logarithmic transformation of the response, motivated by the skewness detected in the exploratory analysis, substantially improved compliance with normality and homoscedasticity, and the use of robust standard errors confirmed the stability of the conclusions.

7. Limitaciones y trabajo futuro

The main limitation of the study lies in the representativeness of the data. The sample covers only male workers from a specific region of the United States between 2003 and 2009, so the conclusions cannot be extrapolated to other populations or to the present context. Added to this is the omission of potentially relevant variables, such as work experience, specific occupation or company size, which would probably explain much of the residual variance.

The data are also cross-sectional, so the estimated coefficients must be interpreted as associations rather than causal effects. This caution is especially necessary in two cases. The coefficient associated with race may be capturing the effect of unobserved variables, such as occupation or differential access to certain sectors, so it should not be read as a direct measure of discrimination. Likewise, the relationship between wage and health insurance is bidirectional, since both variables have been used alternately as predictor and as response in different models.

Several extensions are proposed as lines of future work. First, incorporating interaction terms between education level and job class, which would reveal whether the return on education differs across sectors. Second, replacing the polynomial with splines or generalised additive models, which capture non-linearity more flexibly without

imposing a global functional form. It would also be advisable to replace the single split with k-fold cross-validation, which provides more stable estimates of prediction error, and to explore resampling techniques to address class imbalance in the logistic model. Finally, replicating the analysis on recent data from another geographical setting would make it possible to test the temporal and spatial stability of the results.

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Annex. R Code

```
# =====  
# AG1 - Regresión | Estadística Avanzada (VIU)  
# Autor: Javier Sarmiento Sanjuán  
# =====  
  
# Librerías y datos ----  
library(ISLR2)  
library(ggplot2)  
library(GGally)  
library(car)  
library(lmtest)  
library(caret)  
library(pROC)  
  
data(Wage)  
head(Wage)  
dim(Wage)  
str(Wage)  
  
# ---- 1. Análisis exploratorio ----  
  
# 1.1 Valores perdidos por variable  
colSums(is.na(Wage))  
  
# 1.2 Variable region: comprobar si tiene variabilidad  
table(Wage$region)  
  
# 1.3 Limpieza y preparación  
datos <- Wage  
datos$region <- NULL # un solo nivel con datos: no aporta información  
datos$logwage <- NULL # es log(wage): usarla sería data leakage  
datos$ins <- ifelse(datos$health_ins == "1. Yes", 1, 0) # binaria para la logística  
str(datos)  
  
# 1.4 Resumen estadístico  
summary(datos)  
  
# 1.5 Proporción de la variable binaria  
prop.table(table(datos$ins))  
  
# 1.6 Distribución de la variable respuesta  
ggplot(datos, aes(x = wage)) +  
  geom_histogram(bins = 40, fill = "steelblue", color = "white") +  
  labs(title = "Distribución del salario", x = "Salario (miles USD)", y =  
    "Frecuencia")  
  
# 1.7 Salario vs edad  
ggplot(datos, aes(x = age, y = wage)) +  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "loess", color = "red") +  
  labs(title = "Salario en función de la edad", x = "Edad", y = "Salario")  
  
# 1.8 Salario por nivel educativo  
ggplot(datos, aes(x = education, y = wage)) +  
  geom_boxplot(fill = "lightblue") +  
  labs(title = "Salario por nivel educativo", x = "Educación", y = "Salario") +  
  theme(axis.text.x = element_text(angle = 20, hjust = 1))  
  
# 1.9 Salario por tipo de trabajo  
ggplot(datos, aes(x = jobclass, y = wage)) +  
  geom_boxplot(fill = "lightgreen") +  
  labs(title = "Salario por tipo de trabajo", x = "Tipo de trabajo", y = "Salario")
```

```

# 1.10 Correlaciones entre variables numéricas
cor(datos[, c("year", "age", "wage")])
ggpairs(datos[, c("year", "age", "wage")])

# ---- 2. Regresión lineal / polinómica ----

# 2.1 Regresión lineal simple
m1 <- lm(wage ~ age, data = datos)
summary(m1)

# 2.2 Regresiones polinómicas (grados 2, 3 y 4)
m2 <- lm(wage ~ poly(age, 2), data = datos)
m3 <- lm(wage ~ poly(age, 3), data = datos)
m4 <- lm(wage ~ poly(age, 4), data = datos)

# 2.3 Comparación de modelos anidados (test F)
anova(m1, m2, m3, m4)

# 2.4 Tabla comparativa: R2 ajustado y AIC
modelos <- list(m1, m2, m3, m4)
comparacion <- data.frame(
  grado = 1:4,
  R2_adj = sapply(modelos, function(m) summary(m)$adj.r.squared),
  AIC = sapply(modelos, AIC)
)
comparacion

# 2.5 Curvas ajustadas sobre los datos
ggplot(datos, aes(x = age, y = wage)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", formula = y ~ x, aes(color = "Grado 1"), se = FALSE) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), aes(color = "Grado 2"), se =
FALSE) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 3), aes(color = "Grado 3"), se =
FALSE) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 4), aes(color = "Grado 4"), se =
FALSE) +
  labs(title = "Ajuste lineal vs polinómico", x = "Edad", y = "Salario", color =
"Modelo")

# 2.6 Modelo elegido: grado 3
summary(m3)

# 2.7 Edad con salario máximo estimado
edades <- data.frame(age = 18:80)
edades$pred <- predict(m3, newdata = edades)
edades[which.max(edades$pred), ]

# 2.8 Predicción con intervalos para edades concretas
predict(m3, newdata = data.frame(age = c(25, 45, 65)), interval = "confidence")

# 2.9 Diagnóstico gráfico de residuos
par(mfrow = c(2, 2))
plot(m3)
par(mfrow = c(1, 1))

# 2.10 Tests de supuestos
shapiro.test(residuals(m3)) # normalidad
bptest(m3) # homocedasticidad (Breusch-Pagan)

# 2.11 Transformación logarítmica de la respuesta
m3_log <- lm(log(wage) ~ poly(age, 3), data = datos)
summary(m3_log)
par(mfrow = c(2, 2))
plot(m3_log)

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par(mfrow = c(1, 1))
shapiro.test(residuals(m3_log))
bptest(m3_log)

# ---- 3. Regresión multilíneal ----
library(sandwich)

# 3.1 Modelo completo (respuesta en logaritmo)
mm_full <- lm(log(wage) ~ poly(age, 3) + year + maritl + race + education +
              jobclass + health + health_ins, data = datos)
summary(mm_full)

# 3.2 Multicolinealidad
vif(mm_full)

# 3.3 Selección de variables por AIC (stepwise)
mm_step <- step(mm_full, direction = "both", trace = 0)
formula(mm_step)
summary(mm_step)

# 3.4 ¿Mejora respecto al modelo polinómico simple?
anova(m3_log, mm_step)

# 3.5 Efectos en % sobre el salario
round((exp(coef(mm_step)) - 1) * 100, 1)

# 3.6 Diagnóstico de residuos
par(mfrow = c(2, 2))
plot(mm_step)
par(mfrow = c(1, 1))
shapiro.test(residuals(mm_step))
bptest(mm_step)

# 3.7 Observaciones influyentes
cooks <- cooks.distance(mm_step)
sum(cooks > 4 / nrow(datos))
max(cooks)

# 3.8 Errores estándar robustos (heterocedasticidad)
coeftest(mm_step, vcov = vcovHC(mm_step, type = "HC3"))

# 3.9 Validación train/test (70/30)
set.seed(123)
idx <- createDataPartition(datos$wage, p = 0.7, list = FALSE)
train <- datos[idx, ]
test <- datos[-idx, ]
mm_train <- lm(formula(mm_step), data = train)
pred_test <- exp(predict(mm_train, newdata = test))
c(RMSE = sqrt(mean((test$wage - pred_test)^2)),
  MAE = mean(abs(test$wage - pred_test)),
  R2_test = cor(test$wage, pred_test)^2)

# ---- 4. Regresión logística ----

# 4.1 Variable respuesta como factor (No = 0, Si = 1)
datos$ins_f <- factor(datos$ins, levels = c(0, 1), labels = c("No", "Si"))

# 4.2 Partición train/test estratificada por la respuesta
set.seed(123)
idx2 <- createDataPartition(datos$ins_f, p = 0.7, list = FALSE)
train2 <- datos[idx2, ]
test2 <- datos[-idx2, ]
prop.table(table(train2$ins_f))
prop.table(table(test2$ins_f))

```

```

# 4.3 Modelo nulo y modelo completo
lg_null <- glm(ins_f ~ 1, data = train2, family = binomial)
lg_full <- glm(ins_f ~ age + year + maritl + race + education +
               jobclass + health + wage,
               data = train2, family = binomial)
summary(lg_full)

# 4.4 Selección de variables por AIC
lg_step <- step(lg_full, direction = "both", trace = 0)
formula(lg_step)
summary(lg_step)

# 4.5 Bondad de ajuste
anova(lg_null, lg_step, test = "Chisq")          # test de razón de verosimilitud
c(AIC_null = AIC(lg_null),
  AIC_step = AIC(lg_step),
  R2_McFadden = 1 - lg_step$deviance / lg_step$null.deviance)

# 4.6 Multicolinealidad
vif(lg_step)

# 4.7 Odds ratios con IC 95%
round(cbind(OR = exp(coef(lg_step)), exp(confint.default(lg_step))), 3)

# 4.8 Predicción en test con umbral 0,5
prob_test <- predict(lg_step, newdata = test2, type = "response")
pred_05 <- factor(ifelse(prob_test > 0.5, "Si", "No"), levels = c("No", "Si"))
confusionMatrix(pred_05, test2$ins_f, positive = "Si")

# 4.9 Curva ROC y AUC
roc_obj <- roc(test2$ins_f, prob_test, levels = c("No", "Si"), direction = "<")
auc(roc_obj)
ci.auc(roc_obj)
plot(roc_obj, print.auc = TRUE, asp = NA, axes = FALSE,
     main = "Curva ROC - modelo logístico",
     xlab = "1 - Especificidad", ylab = "Sensibilidad")
axis(1, at = seq(1, 0, -0.2), labels = seq(0, 1, 0.2))
axis(2)

# 4.10 Umbral óptimo (índice de Youden)
umbral <- coords(roc_obj, "best", best.method = "youden")
umbral
pred_opt <- factor(ifelse(prob_test > umbral$threshold[1], "Si", "No"),
                  levels = c("No", "Si"))
confusionMatrix(pred_opt, test2$ins_f, positive = "Si")

```